

Research Briefing | UK and Eurozone

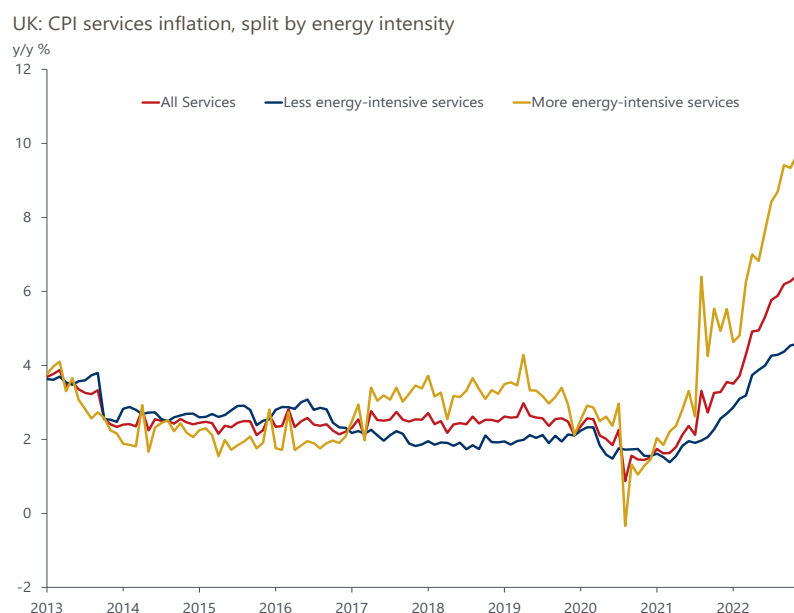
How indirect effects of energy costs are adding to inflation

- The surge in energy prices has significantly lifted headline inflation rates in the UK and Eurozone through the direct impact on household energy bills. Our analysis also suggests that the indirect effects of soaring energy costs for firms have pushed up core inflation considerably, affecting prices for services and non-energy consumer goods in both economies.
- These indirect effects on inflation from energy prices are evident in the large and widening gap between the inflation rates of services and non-energy goods with high energy intensity and those with low energy intensity.
- Because of the lag between changes in wholesale energy prices and their impact on consumer prices, we think the impact on CPI inflation from energy prices will continue to be significant in early 2023 and may even rise further.
- This underscores the risk that core inflation and services inflation will be sticky in the near term. Subsequently, assuming wholesale energy prices follow the path implied by futures markets, the indirect impact on inflation from energy should fade over the next two-three years.
- These indirect effects on inflation from energy prices make it difficult to rely heavily on core inflation and services inflation as precise guides to underlying domestic inflation pressures from capacity use and pay growth. These pressures have risen over the last year, though not as much as implied by the sharp rises in both core and services inflation.

Editor's note: This commentary was authored by Michael Saunders, Oxford Economics' Senior Economic Advisor and former member of the Bank of England's Monetary Policy Committee.

Surging energy prices since early 2021 have lifted inflation directly through increases in prices of petrol, gas and electricity. But they have also had large indirect effects on prices for services and non-energy goods, evident in the wide disparity between the inflation rates of service sector items with high energy intensity and low energy intensity, see **Chart 1**.

Chart 1: UK CPI services inflation has risen much faster among items with high energy use



Source: Oxford Economics, Haver Analytics

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Increases in energy prices affect inflation both directly and indirectly

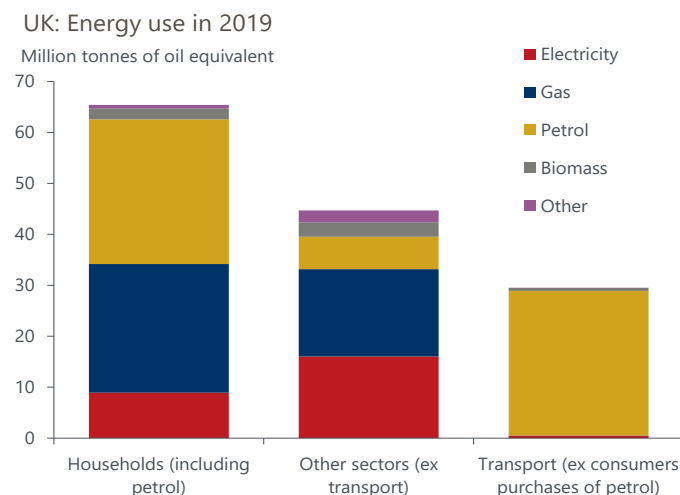
Most discussion of the effects of the surge in energy prices on inflation has focused on the direct effects – notably through increases in prices of petrol, electricity, and gas for households. In the UK, for instance, CPI energy prices, which include household spending on petrol, electricity, gas, and other fuels, and which comprise 6.7% of the CPI, have risen by 53% over the last year. Hence, the direct contribution of the rise in energy prices to CPI inflation has been roughly 3.5ppts (i.e., 6.7% x 53%). Similarly, the direct contribution of higher energy prices to Eurozone inflation has been roughly 2.4ppts over the last year (9.5% weight, 26% YoY increase).

However, the surge in energy prices has also added substantially to CPI inflation through indirect effects that lift the inflation rates of non-energy items. For example, restaurants use a lot of energy to prepare food, and for lighting and heating. The production of the food prepared in restaurants also uses a lot of energy, for example to heat greenhouses and living quarters for animals, fuel for tractors, the transport of food to the restaurant and so forth. So when energy prices rise, restaurants' costs rise and this is likely to be reflected in higher prices for the food and drink sold by restaurants (counted as part of accommodation and food services in the CPI, and part of the service sector).

The story for other non-energy items in the CPI is broadly similar, though the size of the effect varies across sectors because of differences in the amounts of energy that firms use and because their inputs also have different energy intensities.

Estimating these indirect effects on inflation from energy prices is not straightforward, but they are likely to be sizable. After all, in 2019 (latest full data before the pandemic distortions), direct spending on energy by households (i.e., household spending on petrol, electricity, gas, and other fuels) accounted for only about half of total UK energy use, with the rest accounted for by private businesses (including other transport), as well as the public sector.

Chart 2: Roughly half of UK energy use is not directly by households



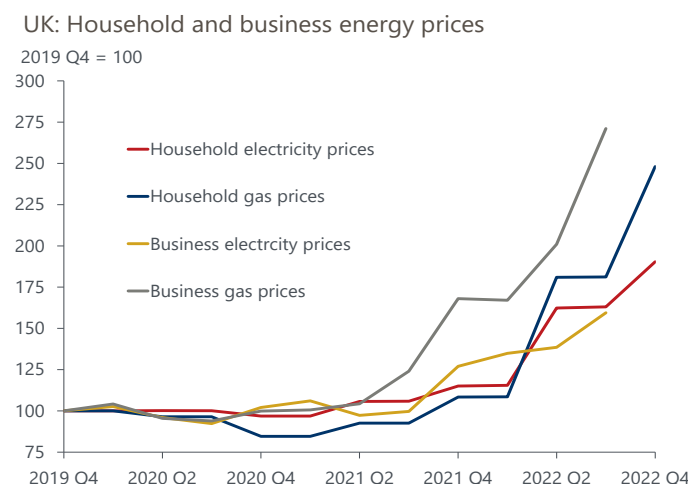
Source: Oxford Economics, Haver Analytics

Moreover, firms' energy costs have risen very rapidly. Indeed, in percentage terms, energy costs rose more for businesses than households over the last year, partly because household bills have higher levels of costs that don't respond to wholesale fuel prices (e.g., network costs, fees, environmental taxes). From Q4 2019 to Q3 2022, average gas prices paid by UK businesses (including the Climate Change Levy) rose by 170%, far above the 80% rise for UK households. The percentage increases in gas prices were even bigger for businesses that use a large amount of gas (because, like households, smaller business users have higher fixed costs).

Household gas prices rose further in Q4 last year – by nearly 40% – bringing the total increase since Q4 2019 to 150%. Official data for business energy prices for Q4 are not yet available but, based on the usual relation with wholesale energy prices, they probably also rose further in Q4 (and would have risen substantially had the government not introduced [the Energy Bill Relief Scheme for businesses](#)).

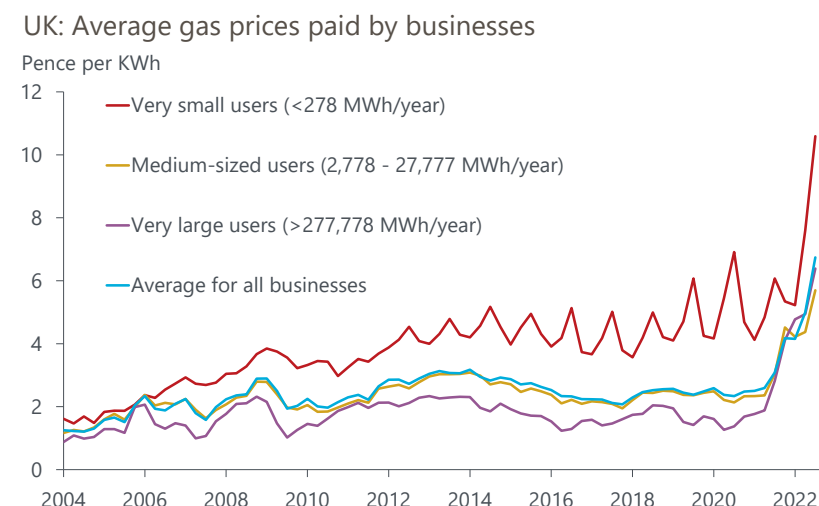
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Chart 3: UK gas prices have risen much faster for businesses than consumers since early 2021



Source: Oxford Economics, Haver Analytics

Chart 4: UK gas prices are higher for small firms, but show a larger percentage rise for intense users



Source: Oxford Economics, ONS

Splitting the CPI into items with high and low levels of energy intensity

For the UK, we used ONS input-output tables to estimate the direct and indirect energy use (including imported energy) as a share of total costs for a detailed disaggregated split of industries. These figures cover the energy that firms use themselves (e.g., for restaurants, this would be the costs for their cooking, heating, and lighting) as well as the energy content of each firm's inputs (e.g., for restaurants, this would include the energy used to produce food ingredients that they buy and then cook). We converted these figures for different industries into estimates for the energy intensity (i.e., energy use as a share of total costs) of consumer spending categories (using [this ONS converter](#)). We then used a detailed split of the CPI to map each individual CPI item onto these consumer spending categories to derive the energy intensity for different CPI items.

The distribution of energy intensity across UK CPI items is highly skewed: among both goods and services, a relatively small number of items have much higher energy intensity than the norm. Among non-energy consumer goods, the categories with highest energy intensity are glassware, tableware, and household utensils; garden plants and flowers; and water supply. The lowest are recording media, books, and newspapers. Among service sector items, energy intensity is of course relatively high among travel services, but also is high for accommodation services (i.e., hotels), as well as restaurants and cafes; canteens; and

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recreation and sporting services. Within the service sector, energy intensity is relatively low for all types of insurance, as well as domestic services and vet services.

In Chart 1, we split UK CPI services items into those with relatively high and relatively low energy intensity (i.e., respectively above or below the average for all CPI services), and show the weighted average YoY inflation rates for both groups. The more energy-intensive items account for about one-third of CPI service sector items by weight (the less energy-intensive items are about two-thirds). Both groups had similar average inflation rates over 2013-2019 (2.5% YoY for less energy-intensive services, 2.7% YoY for more energy-intensive services, and 2.6% for all CPI services).

Since energy prices started to rise in early 2021, inflation has jumped much more among UK CPI services items with high energy intensity, and this has significantly lifted overall CPI services inflation. For example, since 2020, overall CPI services inflation has risen from 1.9% YoY to 6.9% YoY in December 2022 (latest data available), and is well above the 2013-2019 average of 2.6%. Within that, inflation for services that are less energy-intensive has risen from 1.8% YoY in 2020 to 4.6% last December, while the average over 2013-2019 was 2.5% (see [Chart 1](#)). Inflation for services that are more energy-intensive has risen from 2.0% in 2020 to 11.0% in December 2022, and is far above the 2013-2019 average of 2.7%.

Of course, idiosyncratic factors may be affecting inflation for individual items. For example, some services items with high energy intensity are also contact-intensive, and hence may have faced extra cost pressures in reopening after Covid lockdowns. This may have lifted their inflation rates in 2021 and early 2022. Nevertheless, this is less likely to have been a factor in the further surge in inflation for energy-intensive services in the last few quarters, given that Covid reopening was some time earlier. The key difference between these categories is the contribution of energy to their total costs, and it's hard to escape the conclusion that this is the main factor behind the large disparity in inflation rates.

As a rough measure, we take the gap between overall UK CPI services inflation (6.9%) and the inflation rate for UK services that are less energy-intensive (4.6%) as a guide to the indirect effects of energy on services inflation (2.0ppts-2.5ppts at present). This may understate the effects, because the energy price surge has probably also played some role in lifting inflation among services items that are less energy-intensive.

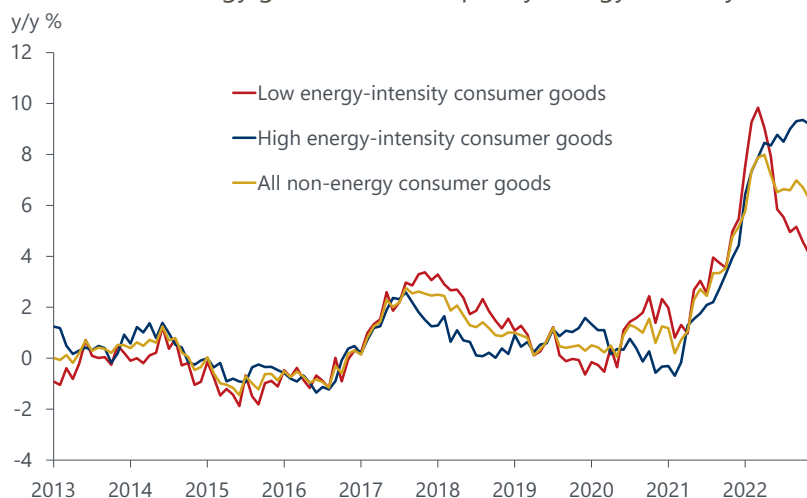
For non-energy consumer goods, a similar exercise shows the average YoY inflation rates for those with relatively low and high energy intensity in the UK CPI (see [Chart 5](#)). The items with high energy intensity account for about 40% of consumer goods. As the chart shows, there was little difference between the inflation rates for these groups in the pre-pandemic period (because swings in energy prices were relatively small). Inflation rose sharply in 2021 for consumer goods with both high and low energy intensity because prices were mainly driven up by widespread supply constraints, rather than energy prices.

Since early 2022, as supply constraints have eased, inflation has slowed sharply for consumer goods with relatively low energy intensity (falling from 9%-10% YoY early last year to 3.3% YoY in December 2022). By contrast, inflation for consumer goods with high energy intensity (8.5% YoY in December) is still slightly higher than in early 2022 (the Q1 2022 average was 7.2% YoY). Using the same approach as before, we judge that gains in energy prices are probably adding about 2.5ppts to non-energy goods price inflation at present (i.e., the gap between inflation for all consumer goods of 5.8% YoY and inflation for consumer goods with relatively low energy intensity of 3.3% YoY).

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Chart 5: Significant gap in inflation between consumer goods of high and low energy intensity

UK: CPI non-energy goods inflation split by energy intensity



Source: Oxford Economics, Haver Analytics

Combined across goods and services, we judge that the indirect effects of higher energy prices may have added about 2.5ppts to UK core inflation over the last year. In other words, had energy prices not surged, core inflation in December 2022 would have been around 3.8% YoY rather than the published figure of 6.3% YoY (ignoring any differences in second round effects, for example on pay and real activity).

Energy costs are probably also adding significantly to inflation for food and soft drinks, which has risen to a multi-decade high of 16.8% YoY (and is not in core inflation). There is not much variation in energy intensity among CPI food items, and all food products in the CPI have energy intensity that is above the average for non-energy goods and services: in other words, they are all relatively energy-intensive.

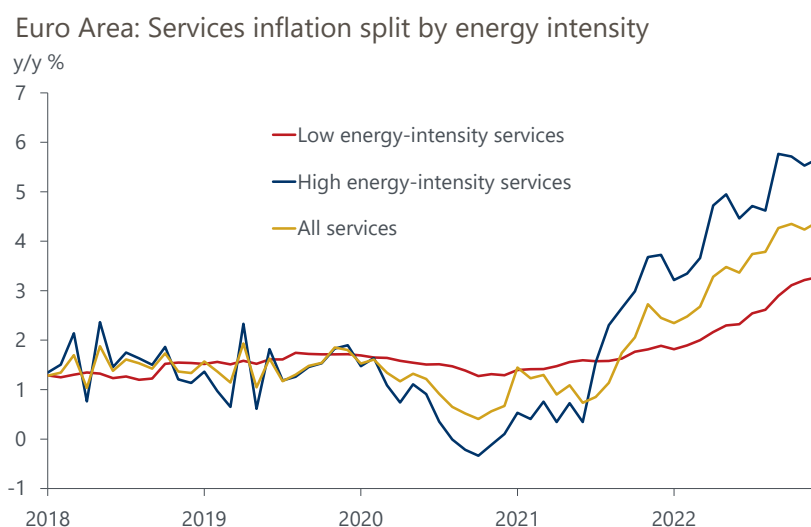
Indirect effects on inflation also are large in the Eurozone

The indirect effects on inflation from higher energy prices also appear to be large in the Eurozone. As with the UK, energy prices have risen faster for businesses than consumers, partly because of the fixed costs discussed earlier. In H1 2022 (latest full data), average gas prices paid by Eurozone households were up about 50% YoY, whereas gas prices for mid-sized non-households (i.e., business users) were up about 125% YoY (with, as in the UK, bigger percentage increases for large energy users).

Charts 6 and 7 show a similar split of Eurozone services inflation and non-energy goods inflation between items that are more or less energy intensive, using the definitions outlined in a [blog by ECB Executive Board Member Philip Lane](#). Using the ECB definitions, energy intensive items account for just over 40% of total services in the CPI by weight and just below 40% of non-energy goods in the CPI. As with the UK, there was little difference between the average inflation rates for items with low and high energy intensity in the period before the pandemic, but a wide gap has emerged in the last year. Overall CPI services inflation has risen from 1.5% YoY in 2019 to 4.4% in December 2022 – a rise of nearly 3ppts. Within that, inflation for services items with low energy intensity has risen from 1.6% YoY in 2019 to 3.3% now (rise of 1.7ppts), while inflation for services items with high energy intensity has risen from 1.4% in 2019 to 5.7% now (rise of 4.3ppts).

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Chart 6: Eurozone services inflation has been pushed up by energy prices

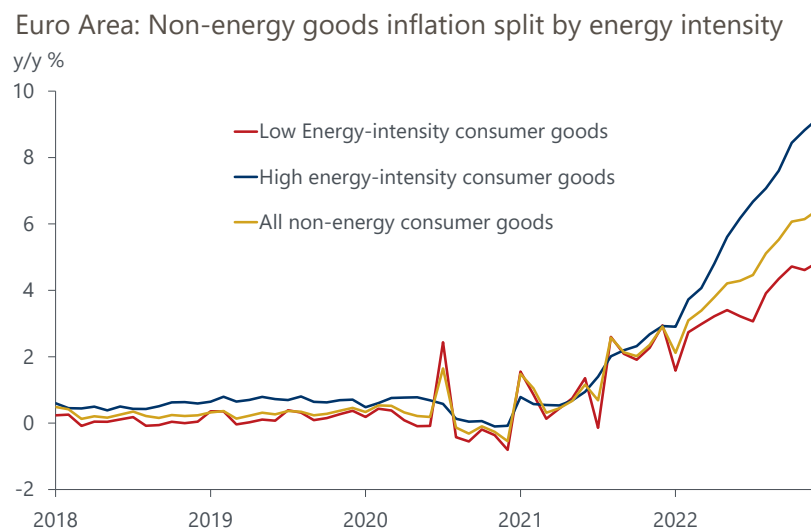


Source: Oxford Economics, Haver Analytics

Among all non-energy consumer goods, inflation has risen from 0.3% YoY in 2019 to 6.4% in December 2022 (rise of just over 6ppts) in the Eurozone. Within that, inflation has risen from 0.2% to 4.8% among items with relatively low energy intensity (rise of 4.6ppts), and from 0.7% to 9.2% among items with high energy intensity (rise of 8.5ppts).

Using this as a rough guide, it's likely that energy prices have added 1ppt-2ppts to both goods and services inflation in the Eurozone over the last year (with a similar effect on core inflation).

Chart 7: Eurozone goods inflation also much higher among energy-intensive items



Source: Oxford Economics, Haver Analytics

We stress that these estimates of the indirect effects of energy prices on inflation in the UK and Eurozone are somewhat uncertain. One could argue for a bigger effect (by including some of the rise in inflation among items with low energy intensity). Moreover, the passthrough may have been affected by government support for some businesses. We would not make too much of the point that our analysis produces slightly smaller indirect effects on inflation in the Eurozone than in the UK, given that we use slightly different splits of CPI items into more or less energy-intensity categories, and that government support packages differ across countries. Rather than the precise figures, the key point is that these indirect effects on inflation appear to have been quite large in both economies and have increased recently.

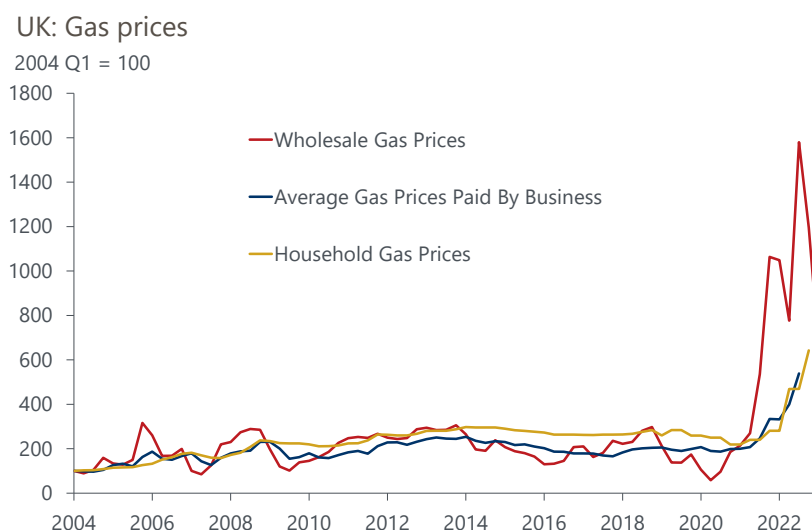
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Indirect effects on inflation are likely to remain large in early 2023

There are lags between changes in wholesale energy prices and the subsequent indirect effects on CPI inflation. These lags result from two factors.

First, business energy prices typically do not respond immediately to changes in wholesale energy prices, because many firms have contracts that fix prices for a period. The ONS BICS survey suggests that in the UK, 29% of firms (excluding the 'don't knows' and firms with less than 10 employees) have variable price contracts for gas, 13% have prices fixed for one quarter ahead, 13% have prices fixed for two quarters ahead, and roughly 45% have prices fixed for more than six months. Hence, even allowing for the fixed cost element in firms' gas prices, the average prices charged for business gas (and electricity) prices in recent quarters did not fully reflect the surge in wholesale energy prices over that period. For example, as noted, average gas prices for businesses rose by about 170% between Q4 2019 and Q3 2022, but that rise was a long way short of the surge in wholesale gas prices over that period (one-month forward contract for UK gas prices rose by 800% between Q4 2019 and Q3 2022, and was nearly 600% above the Q4 2019 level in Q4 2022).

Chart 8: The recent drop in wholesale gas prices is unlikely to produce an early decline in average gas prices for businesses



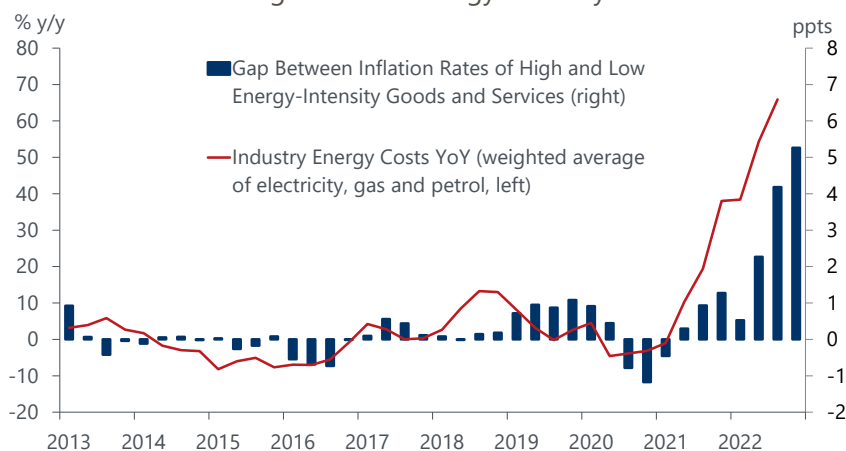
Source: Oxford Economics, Haver Analytics

The second factor is that the passthrough of firms' energy costs to consumer prices does not seem to be immediate. For example, the gap between the inflation rates of items (consumer goods and services) with high energy intensity and low energy intensity tends to respond over a few quarters to changes in the average energy prices paid by businesses.

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Chart 9: Swings in firms' energy costs usually feed through to consumer prices over several quarters

UK: Changes in business energy costs and gap between inflation rates of items with high and low energy intensity



Source: Oxford Economics, Haver Analytics

Given these lags, it's unlikely that CPI prices in the UK and Eurozone have already fully responded to the surge in wholesale gas prices late last year. Further effects from those increases are likely in coming months.

Wholesale energy prices have fallen recently, but the one-month forward gas price is still about 275% above the Q4 2019 level. It follows that there probably will still be some catch-up increases in business energy costs during this year as fixed-price contracts for gas and electricity expire and are repriced at current levels. In the UK, the level of government support for firms' energy costs will diminish markedly after the [Energy Bill Relief Scheme](#) ends and the [Energy Bills Discount Scheme](#) begins in April. If wholesale gas prices stabilise around current levels (in line with futures prices), then the YoY increase in business energy costs will subside markedly during 2023. But this slowdown will mainly occur in H2 rather than H1. In addition, the surge in business energy costs in late 2022 and early 2023 will continue to feed through to consumer prices, especially for items with relatively high energy intensity.

As a result, the indirect effects on consumer price inflation from energy prices (on both services and non-energy goods) probably will remain large in early 2023, similar to (or perhaps a bit greater than) the effects in Q4 2022.

This is likely to add to stickiness in overall (and core) CPI inflation in both the UK and Eurozone in early 2023, despite likely weakness in real activity and the easing of supply-side bottlenecks that previously lifted some goods prices. Thereafter, assuming wholesale energy prices follow the path implied by futures markets, these indirect effects on inflation should start to fall in late 2023 and then fade away during 2024-2025.

Implications: Inflation is a complicated, nuanced picture

First, while it may be tempting to regard core inflation – and especially services inflation – as relatively 'pure' measures of inflation (i.e., linked to domestic capacity pressures rather than energy costs), in practice these measures have also been significantly increased in the UK and eurozone by indirect effects from the energy price surge. As guides to underlying inflation trends, it's better to focus on core and services inflation rather than headline inflation; but it is better still to look at measures of inflation among items with relatively low energy intensity.

Second, while gains in energy prices can explain part of the rise in services inflation and core inflation over the last 12-18 months, they cannot explain all of it. There also has been a significant rise in underlying inflation pressures, consistent with tight labour markets and higher pay growth.

Third, despite the recent declines in wholesale energy prices, the indirect effects on inflation from last year's gains in wholesale energy prices probably will remain large, and may even rise further in early 2023. This reflects lags in the adjustment of business energy costs to wholesale prices, as well as lags between

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changes in business energy costs and consumer prices. These indirect energy effects on inflation reinforce the risk that core inflation and services inflation will be quite sticky near term, despite weak demand and easing supply bottlenecks.

Fourth, assuming energy prices follow the path implied by futures markets, these large indirect effects on inflation from energy prices will eventually fade away – most likely from H2 2023 and during 2024-2025 – and are unlikely to depend heavily on demand and capacity pressures in the economy. But it is likely to add significantly to downward effects on inflation in that period.

All this is likely to pose significant challenges for the BoE and ECB. Both central banks have, understandably, argued that the direct effects of energy prices on inflation are likely to fade over time. As a result, in setting monetary policy and explaining their decisions, they have to an extent downplayed the rise in headline inflation rates. Instead, they have put increasing emphasis on core inflation and services inflation as cleaner guides to underlying – and potentially more persistent – inflation pressures that are linked to domestic capacity pressures and pay growth rather than energy prices. In turn, the marked increases in services inflation and core inflation have added to the greater hawkishness of both central banks since mid-2022.

However, our analysis suggests that the recent strength in both core and services inflation measures partly reflects significant indirect effects from higher energy prices (a factor that is unlikely to persist at the policy-relevant horizon of 2-3 years ahead). This does not mean that core and services inflation have no value as guides to underlying inflation trends. Moreover, energy prices cannot fully account for the recent rise in services inflation or core inflation. And it's worth noting that labour markets in the UK and Eurozone are tight and that pay growth has also picked up, especially in the UK. But our analysis does suggest that core and services inflation cannot be relied on too heavily as definitive guides to underlying (i.e., non-energy) inflation pressures.

We do not believe that the BoE and ECB can credibly adopt a stance of downplaying all inflation gauges, given that they target inflation. But they may adopt a more nuanced approach. For example, they may seek to highlight indirect effects from energy prices on core and services inflation to better gauge the extent of inflation pressures that are potentially persistent (and hence relevant for monetary policy). And, as well as seeking cleaner measures of underlying inflation, they may put a high emphasis on labour market data, notably unemployment and under-employment as guides to capacity use, and pay growth as a guide to underlying cost pressures.